

SPC Nelson Rules - 8 Out-of-Control Signals

When to take action on your control chart | AIAG SPC 2nd Edition

QUALITY HEAD AGENTS | AIAG SPC 2nd Edition | IATF 16949:2016

Nelson Rules - The 8 Out-of-Control Signals

- Nelson Rules (also called Western Electric Rules) define when a control chart shows special cause variation.
- Any of these 8 signals means: STOP, CONTAIN, INVESTIGATE, CORRECT.
- All operators on SPC lines must be trained to identify and respond to all 8 rules.
- Customer requirement: define which rules are applied and show training evidence.

Rules 1-4 (Most Common)

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| Rule 1: | 1 point beyond 3-sigma (UCL or LCL). Most obvious signal. Stop immediately. |
| Rule 2: | 9 consecutive points on same side of centerline. Process has shifted. Investigate mean shift. |
| Rule 3: | 6 consecutive points trending up or down. Tool wear, fixture wear, temperature drift. |
| Rule 4: | 14 consecutive points alternating up/down. Over-adjustment - operator tampering. |

Rules 5-8 (Less Common but Important)

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| Rule 5: | 2 of 3 consecutive points beyond 2-sigma on same side. Increasing variation signal. |
| Rule 6: | 4 of 5 consecutive points beyond 1-sigma on same side. Subtle process shift. |
| Rule 7: | 15 consecutive points within 1-sigma of centerline. Stratification - data from multiple streams. |
| Rule 8: | 8 consecutive points beyond 1-sigma on either side with none in the middle. Bimodal process. |

Reaction to Out-of-Control Signal

- Step 1: Stop the machine. Do not continue production.
- Step 2: Identify and segregate all parts produced since the last in-control signal.
- Step 3: 100% inspect or sort segregated parts per reaction plan.
- Step 4: Identify root cause of the signal. Record in the out-of-control log.
- Step 5: Take corrective action. Restart only after process is verified stable.
- Step 6: Record the event on the control chart (note, date, action taken).
- Auditor Evidence: Out-of-control log entries showing date, rule triggered, action taken, restart verification.